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Conceptual Framework for Deepfake Audio Detection in Call Centers Using Spectral and Temporal Analysis

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Abstract

The rapid advancement of generative artificial intelligence has enabled highly realistic synthetic speech, posing critical security threats to voice-based authentication systems in call-center environments. Existing deepfake audio detection approaches achieve strong performance in controlled laboratory settings but fail under real-world telephony conditions including codec compression, background noise, packet loss, and bandwidth limitations.

This study proposes a conceptual framework for deepfake audio detection using hybrid spectral and temporal analysis specifically designed for call-center deployment. The framework comprises six interconnected modules: telecommunication preprocessing and normalization, hybrid spectral-temporal feature extraction, dual-stream CNN-BiLSTM analysis engine, adaptive decision fusion with Bayesian belief updating, output interpretation and fraud alerting, and real-time deployment architecture targeting sub-100-ms latency.

The system addresses telephony degradation through integrated codec normalization, noise simulation, and packet-loss handling. Comparative evaluation against nine representative detection systems demonstrates unique capabilities in spectral-temporal fusion, telephony robustness, and real-time feasibility.

Implementation specifications include Python-based development, ASVspoof 2019/2021 benchmark evaluation, and comprehensive metrics including accuracy, Equal Error Rate, and inference latency.

Keywords: Deepfake audio detection, voice spoofing, spectral analysis, temporal analysis, call-center security, CNN-BiLSTM, telecommunication robustness.

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Chapter 1

INTRODUCTION

INTRODUCTION

1.1 Research Background

The rapid evolution of deep learning techniques has given rise to new and complex challenges in the manipulation of digital media. Image, audio, and video files are increasingly being manipulated using artificial intelligence-based methods known as “deepfake”, a portmanteau of deep learning and fake [1][2]. The term “deepfake” first entered the public domain in 2017, when a Reddit user with the username “deepfakes” shared manipulated videos online. These videos were created using Google’s open-source deep learning technology to swap faces in existing footage, demonstrating the potential of AI-driven synthetic media [3].

Between 2018 and 2019, there was a significant increase in the number of articles about deepfake (from 60 to 309) [4]. Articles about deepfakes were expected to increase to over 730 by 2020’s end, according to predictions made on July 24 [5]. In [6] found that most focus is on video deepfakes, particularly in developing video deepfakes.

Deepfake refers to a form of digitally created content in which artificial intelligence techniques are used to manipulate or generate audio, video, or images, often by replacing or imitating the appearance or voice of real individuals. Among these applications, the use of deepfake techniques in audio is particularly concerning, as it enables the creation and distribution of highly realistic but fake speech recordings. This raises serious concerns regarding the reliability, authenticity, and security of audio- based communication systems [7].

There are several approaches to creating deepfake audio, with the three most commonly used techniques being text-to-speech (TTS), voice conversion (VC), and replay attacks [7].

Text to Speech TTS (speech synthesis (SS))

Text to speech is a technique that generates artificial speech directly from textual input using deep learning models. Modern TTS systems are capable of producing highly natural and human-like voices, often mimicking specific individuals by learning their vocal characteristics. These systems typically rely on neural architectures such as Tacotron and WaveNet, which model both linguistic and acoustic features to synthesize realistic speech. In malicious contexts, TTS can be used to create convincing fake audio without requiring the attacker to speak [7][8][9].

Figure 1.1 illustrates the workflow of Text-to-Speech (TTS) deepfake generation, where textual input is converted into synthetic speech using deep learning models trained on a target speaker’s voice.

Text-to-Speech (TTS) Deepfake Audio – How it Occurs

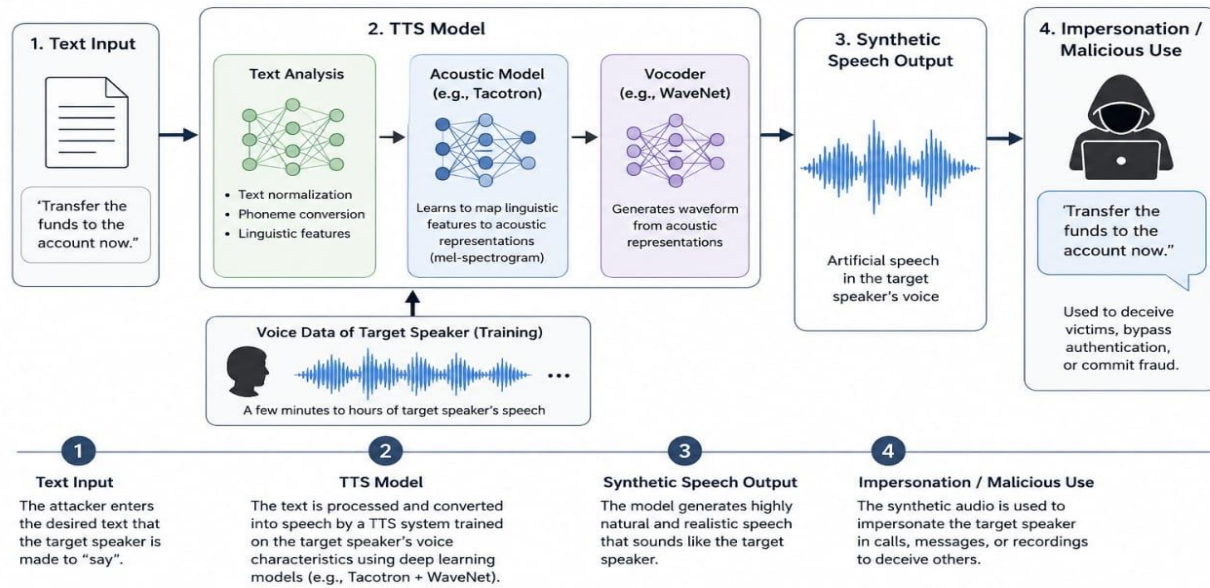


Figure 1.1. Workflow of Text-to-Speech (TTS) deepfake audio generation

Voice Conversion VC

Voice conversion involves transforming the voice of a source speaker to sound like that of a target speaker while preserving the original linguistic content. Unlike text to speech, which generates speech from text, voice conversion modifies existing speech by altering acoustic features such as pitch, timbre, and speaking style. This technique is particularly dangerous in real time scenarios, as attackers can speak naturally while their voice is converted to impersonate another individual during live interactions [7].

Figure 1.2 presents the process of Voice Conversion (VC), in which an attacker's speech is transformed to mimic the target speaker while preserving the original linguistic content.

Voice Conversion (VC) Deepfake Audio – How it Occurs

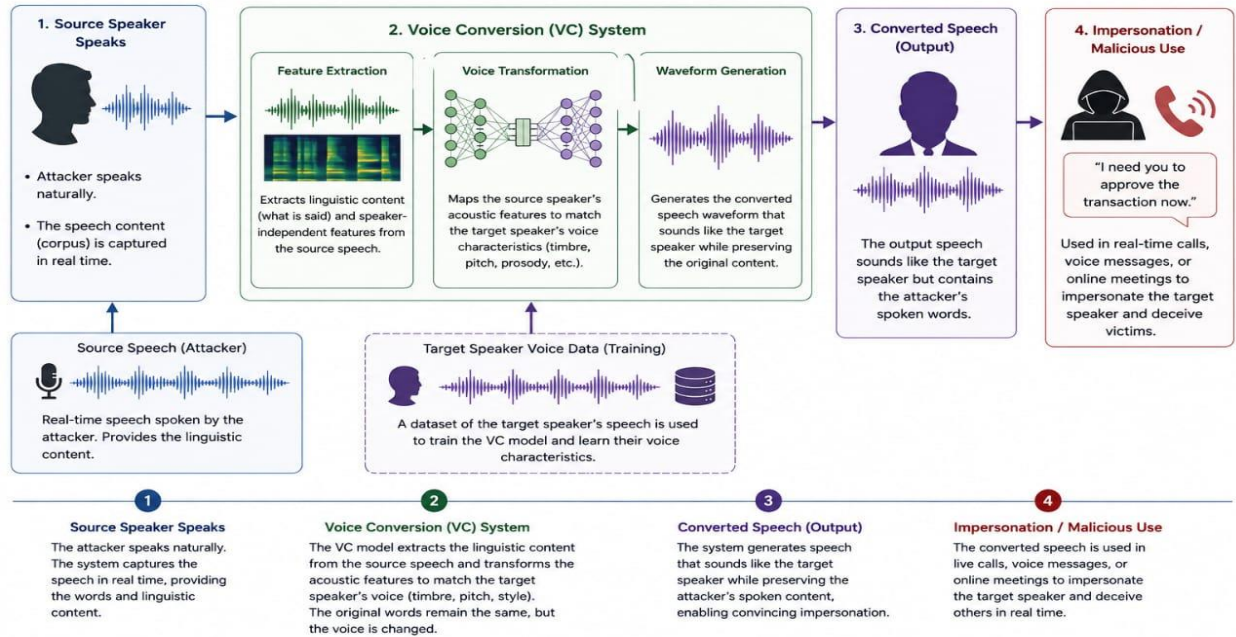


Figure 1.2. Workflow of Voice Conversion (VC) deepfake audio generation.

Replay Attacks

Replay attacks involve the use of pre-recorded genuine speech to impersonate a legitimate speaker. In this approach, an attacker records a target individual's voice and replays it to bypass authentication systems. Unlike text to speech and voice conversion, replay attacks do not involve synthetic generation or transformation of speech but instead rely on authentic recordings. However, they may still include channel distortions such as background noise or compression artifacts introduced during recording and playback [10][11][12].

Figure 1.3 demonstrates the replay attack process, where previously recorded genuine speech is replayed to impersonate a legitimate speaker or bypass authentication systems.

Replay Attack Deepfake Audio – How it Occurs

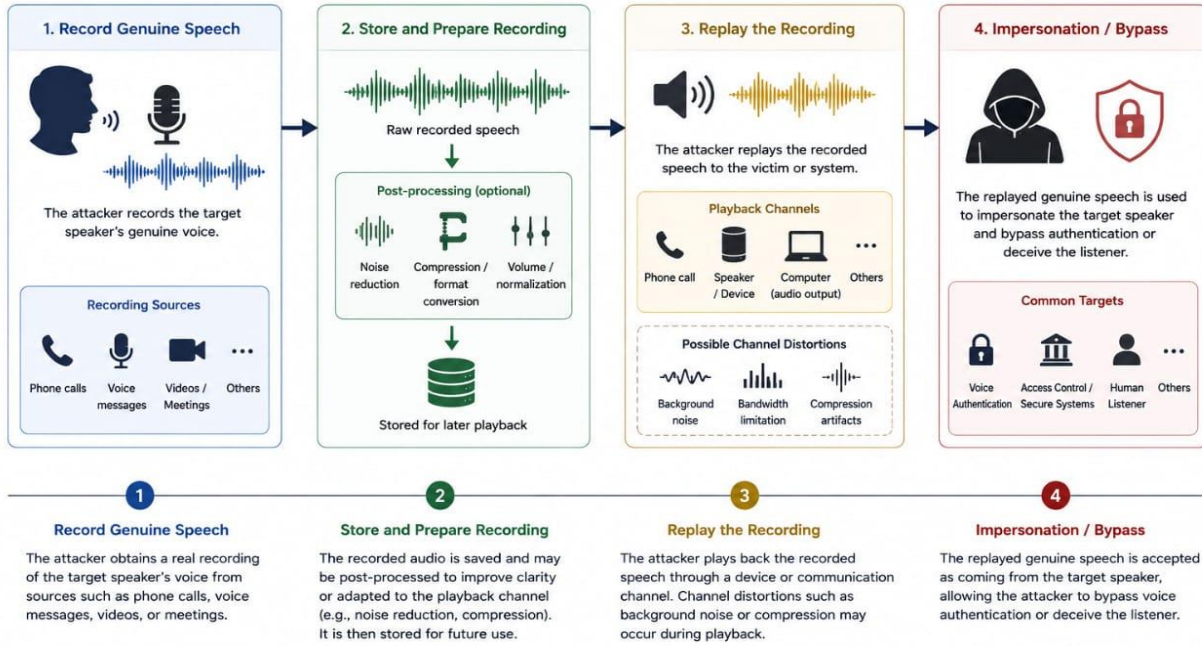


Figure 1.3. Workflow of replay attack deepfake audio impersonation.

The rapid advancement of generative artificial intelligence has created an unprecedented challenge for voice-based authentication systems, particularly in high stakes call center environments. Voice authentication has become a critical biometric method for financial institutions, telecommunications providers, and customer service operations, supporting millions of daily interactions that require reliable identity verification [13]. However, the emergence of advanced text to speech and voice conversion technologies has exposed traditional voice biometric systems to synthetic speech attacks that can convincingly imitate legitimate users.

Call centers have become attractive targets for deepfake enabled fraud due to their reliance on audio-based authentication processes. Modern attackers employ multi vector strategies that combine phishing, credential harvesting, and AI driven voice synthesis to carry out account takeovers and unauthorized transactions. The financial implications are significant, with a large proportion of account takeover incidents in financial services originating from call center interactions. Traditional voice biometric systems analyze vocal characteristics such as pitch patterns, speaking rhythm, and frequency distribution to generate voiceprints, yet these systems lack mechanisms to verify whether the speech itself is genuine or artificially generated.

In response to these threats, deepfake detection methods have evolved toward architectures that analyze both spectral and temporal characteristics of speech signals. Spectro temporal features, including Linear Frequency Cepstral Coefficients, Mel Frequency Cepstral Coefficients, Constant Q Cepstral Coefficients, and log power spectrograms, capture frequency-based patterns and short-

term variations that reveal artifacts introduced during speech synthesis [14][15][16][17][18]. At the same time, temporal analysis focuses on speech dynamics such as pause patterns, rhythm consistency, and prosodic variations, which often expose unnatural patterns in synthetic audio. The integration of these complementary approaches has demonstrated improved performance in detecting previously unseen spoofing attacks.

Recent advances in self-supervised learning have further enhanced deepfake audio detection capabilities [19]. Models such as WavLM, Wav2Vec 2.0, and XLS R learn robust speech representations from large scale unlabeled data. In particular, WavLM incorporates speech denoising and prediction tasks, enabling it to capture both speaker specific attributes and environmental characteristics that are useful for detecting synthetic artifacts. When combined with efficient classification models, these approaches achieve high detection accuracy while maintaining computational efficiency suitable for real time applications.

Despite these advancements, call center environments impose strict operational constraints on detection systems. Real time voice authentication requires processing within a few seconds without interrupting natural conversation flow, demanding a balance between detection accuracy and computational efficiency. In addition, telephony conditions such as limited sampling rates, compression artifacts, and transmission noise introduce distortions that can obscure synthetic speech characteristics. These challenges highlight the need for robust and efficient detection frameworks tailored specifically for real time call center deployment.

1.2 Problem Statement

Despite significant advances in deepfake audio detection, a critical gap persists between research-oriented systems and deployable call center solutions. Existing detection frameworks are predominantly developed as standalone laboratory models, optimized for clean and high-fidelity audio conditions, with limited consideration for real-world telephony environments. As a result, they lack effective integration with telephony infrastructure and real-time authentication workflows. This disconnect creates a vulnerability window in which sophisticated synthetic speech attacks can bypass security mechanisms before detection occurs.

The key challenges addressed in this research include:

Generalization Deficit:

Current detection models often experience significant performance degradation when exposed to unseen synthesis techniques or cross-domain conditions, particularly those involving telephony distortions. The rapid advancement of text-to-speech (TTS) and voice conversion (VC) technologies necessitates the development of adaptive detection systems capable of identifying emerging attack methods without requiring frequent retraining.

Real-time Processing Constraints:

Call center authentication systems require near-instantaneous decision-making within live conversational settings. However, many high-performing detection models rely on computationally intensive architectures that fail to meet strict latency requirements. This creates a critical trade-off between detection accuracy and processing efficiency, limiting their practical deployment.

Environmental Robustness:

Telephony channels introduce various distortions, including compression, packet loss, and background noise, which can obscure the distinguishing features of synthetic speech. These conditions often result in overlapping feature distributions between genuine and spoofed audio, making accurate classification more challenging.

Integration Complexity:

Most existing detection systems lack standardized interfaces, such as APIs or microservices architectures, required for seamless integration with operational platforms like interactive voice response (IVR) systems, customer relationship management (CRM) tools, and fraud detection pipelines. This limits their usability in real-world applications.

In light of these challenges, the fundamental problem addressed by this research is the lack of a unified conceptual framework that integrates spectral and temporal analysis techniques within a real-time, call center-oriented architecture. Such a framework must support adaptive learning, robust performance under telephony conditions, and seamless integration into existing operational systems.

1.3 Research Questions

This study seeks to address the following research questions:

RQ1. What are the limitations of existing deepfake audio detection approaches in call center environments?

RQ2. How can spectral and temporal analysis techniques be effectively combined within a conceptual framework for detecting deepfake audio?

RQ3. What key features (spectral and temporal) are most suitable for identifying voice spoofing attacks in telephony-based systems?

RQ4. How can a conceptual framework be designed to support real-time detection of deepfake audio in call center environments?

RQ5. How can the proposed framework be structured to ensure robustness against telephony channel distortions and emerging deepfake techniques?

1.4 Research Aims and Objectives

Overall Purpose of the Study

The overall purpose of this study is to develop a conceptual framework for detecting deepfake audio in call center environments using spectral and temporal analysis techniques, ensuring robustness, real-time capability, and operational feasibility.

Aims

- 1.To design a conceptual detection framework that leverages spectral and temporal features to identify deepfake audio in call center communications.
- 2.To propose an operational integration model that enables real-time deployment within existing call center infrastructures.

Objectives

- 1.To **review** existing deepfake audio detection techniques and identify their limitations in call center environments.
2. To **examine** spectral and temporal feature extraction methods relevant to detecting synthetic speech.
- 3.To **design** a conceptual framework that integrates spectral and temporal analysis for deepfake audio detection.
4. To **define** the key components and processes involved in the proposed detection framework.
- 5.To **evaluate** how the proposed framework can support real-time detection in telephony-based systems.
- 6.To **propose** strategies for ensuring robustness and adaptability against evolving deepfake audio techniques.

1.5 Research Timeline

The research will be carried out over an approximate four and a half-month period, spanning from January to 15th May 2026. This timeline is structured to ensure a systematic and well-organized progression of all research activities, from initial review to final submission. The detailed schedule is presented below:

Task Name	Start Date	Finish Date	Duration
Research Planning & Topic Refinement	01 Jan 2026	10 Jan 2026	10 Days
Chapter Two: Literature Review	11 Jan 2026	31 Jan 2026	21 Days
Chapter One: Introduction	01 Feb 2026	10 Feb 2026	10 Days
Chapter Three: Methodology & Framework Development	11 Feb 2026	31 Mar 2026	49 Days
Proposed Implementation	01 Apr 2026	25 Apr 2026	25 Days
Chapter Four: Results & Discussion	26 Apr 2026	05 May 2026	10 Days
Revision & Final Editing	06 May 2026	12 May 2026	07 Days
Final Submission	13 May 2026	15 May 2026	03 Days

Table 1.1: Research Timeline.

In addition, the research timeline is presented using a Gantt chart to provide a clear visual representation of the sequence, duration, and overlap of the various research activities.

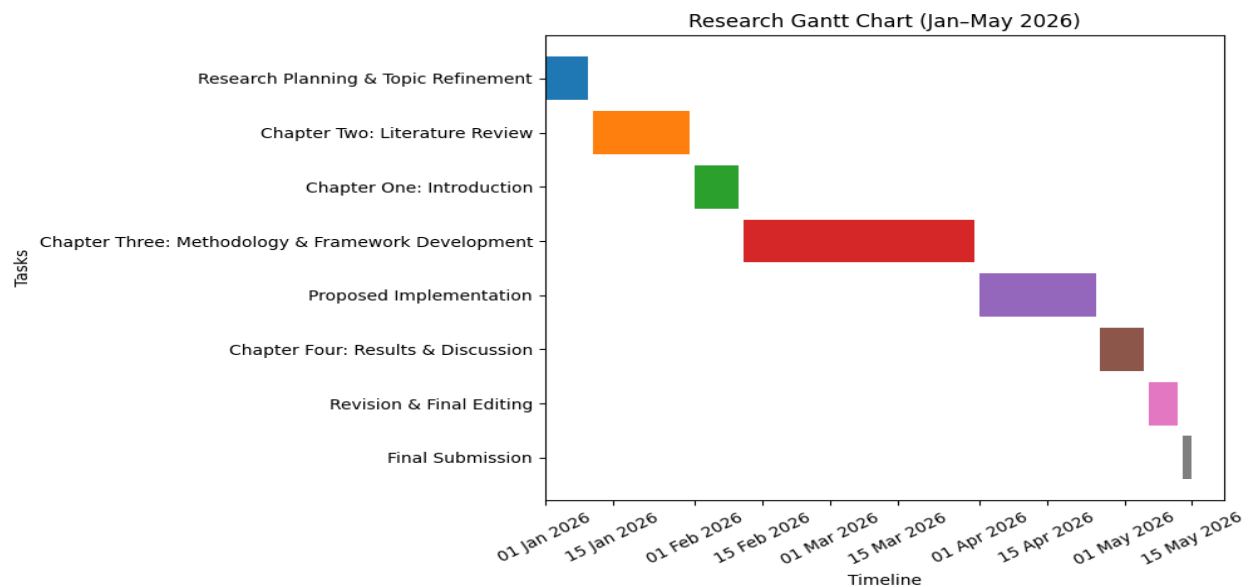


Figure 1.4: Research Timeline Gantt Chart

1.6 Research Organization

This research is organized into five chapters, each addressing a specific aspect of the study as follows:

Chapter 1: Introduction

This chapter presents the background of the study on deepfake audio threats to call-center voice authentication systems, covering text-to-speech synthesis, voice conversion, and replay attacks. It states the problem regarding the gap between laboratory detection models and real-world telephony conditions, presents the research questions, and outlines the aims, objectives, research timeline, and overall organization of the research.

Chapter 2: Literature Review

This chapter reviews existing literature on deepfake audio generation and detection techniques. It discusses previous studies on spectral and temporal analysis methods, machine learning and deep learning approaches, transfer learning, lightweight architectures, and hybrid models. The chapter evaluates datasets, performance metrics, and current challenges, and identifies the research gap concerning telephony robustness, real-time deployment, and spectral-temporal integration.

Chapter 3: Methodology and Proposed Framework

This chapter describes the quantitative, experimental research methodology adopted in the study, including the research design, data preparation, feature extraction techniques, and model architecture. It presents the proposed six-module conceptual framework for deepfake audio detection in call center environments, explaining the telecommunication preprocessing pipeline, hybrid spectral-temporal feature extraction, dual-stream CNN-BiLSTM analysis engine, adaptive decision fusion, and real-time deployment architecture.

Chapter 4: Results and Discussion

This chapter presents the comparative evaluation and discussion of the proposed framework against nine related studies. It analyzes the framework's spectral-temporal integration, telephony robustness, computational efficiency, generalization capability, and explainability, assessing its strengths, adaptability, and deployment suitability within real-world telephony environments.

Chapter 5: Conclusions and Future Work

This chapter summarizes the major findings and contributions of the study. It discusses the limitations of the proposed framework, outlines strategies for ensuring robustness and adaptability against evolving deepfake techniques, and provides recommendations and future research directions for practical implementation, full-scale deployment, and further development.

Chapter 2

LITERATURE REVIEW

2.1 Introduction

The rapid advancement of artificial intelligence and deep learning has significantly increased the sophistication of deepfake audio generation techniques, making the detection of synthetic and manipulated speech a major research challenge in security-sensitive environments. Call centers, voice authentication systems, and financial service hotlines are particularly vulnerable, as deepfake audio attacks, including Text-to-Speech synthesis, Voice Conversion, and replay-based spoofing can imitate human speech with near-perfect realism. These attacks bypass traditional voice verification systems that rely on static acoustic features, creating an urgent need for more robust detection methodologies.

To address these threats, researchers have proposed a wide spectrum of machine learning and deep learning approaches for voice spoofing detection. Existing studies have explored techniques grounded in spectral analysis, temporal dependency modeling, convolutional neural networks, recurrent neural networks, attention mechanisms, and hybrid architectures [20]. The core objective across these methods is to identify subtle inconsistencies and microscopic artifacts within synthetic speech signals, such as spectral phase mismatches, unnatural temporal transitions, and missing prosodic variations that distinguish them from genuine human speech.

This chapter reviews existing studies related to deepfake audio and voice spoofing detection techniques. The reviewed studies are evaluated in terms of their architectural choices, feature extraction strategies, benchmark datasets, performance metrics, and operational limitations. A comparative analysis is provided to highlight methodological trends, performance trade-offs, and deployment barriers. Finally, this chapter identifies the research gap that motivates the proposed framework in this study.

2.2 Previous Studies on Deepfake Audio Detection

The detection of synthetic and manipulated speech has emerged as a critical research area in digital forensics and biometric security. Early approaches relied primarily on handcrafted acoustic features such as Mel-Frequency Cepstral Coefficients, Linear Predictive Coding, and spectral centroid combined with traditional machine learning classifiers including Support Vector Machines and Gaussian Mixture Models. While effective against rudimentary spoofing attempts, these methods struggle to generalize against modern deepfake audio generated by advanced neural architectures. The rapid evolution of text-to-speech and voice conversion technologies has necessitated the development of more sophisticated deep learning-based detection frameworks capable of jointly capturing complex spectral textures and long-term temporal dependencies inherent in synthetic speech.

The following subsections review prominent studies in deepfake audio detection, organized from the most recent advances to foundational contributions. Each study is examined in terms of its

architectural design, feature representation strategy, evaluation methodology, and inherent limitations.

2.2.1 Transfer Learning and Hybrid Architectures

Shahzad et al. [21] introduced a hybrid VGGish-LSTM model to address voice spoofing attacks targeting Automatic Speaker Verification systems. The approach utilizes transfer learning-based embeddings extracted via the VGGish model to capture both spectral and temporal features, which are then processed by an LSTM network to model long-term dependencies in speech. Evaluated on the ASVspoof 2019 Logical Access dataset, the model achieved an accuracy of 99.11% and an Equal Error Rate of 1.18%, outperforming CNN-based and YAMNet-based alternatives. Despite its strong performance, the study highlighted limitations such as dataset dependency, potential overfitting, and limited coverage of emerging deepfake attack types [21].

Verma et al. [22] conducted a comparative study of various deep learning models, including Bidirectional Long Short-Term Memory and Convolutional Neural Networks, for deepfake audio detection. The models were evaluated using both time-domain features such as Zero Crossing Rate and Root Mean Square Energy and spectral features including Mel-Frequency Cepstral Coefficients extracted from the ASVspoof 2019 dataset. The best-performing model, a Squeeze-and-Excitation Enhanced 1D CNN, achieved 97.64% accuracy, along with high precision, recall, and F1-score values of approximately 98%. However, the study noted several limitations, including dataset dependency, lack of diversity in accents and speech styles, limited evaluation of complex attack types, and high computational cost, which may hinder real-world deployment [22].

Shah [23] proposed a comprehensive AI-driven framework for fraud detection across mobile communication channels, including SMS, phone calls, and messaging platforms. The study employed deep learning architectures such as Long Short-Term Memory and hybrid CNN-LSTM models, alongside acoustic features like Mel-Frequency Cepstral Coefficients and spectral analysis to detect AI-generated voice scams. The model achieved high accuracy levels ranging from 94% to 99% across different tasks. The study also emphasized the importance of multi-layered detection systems that integrate real-time anomaly detection with behavioral analysis to address increasingly sophisticated deepfake-based attacks [23].

2.2.2 Lightweight and Spectral-Focused Architectures

Kwak et al. [24] proposed a lightweight voice spoofing detection architecture known as ResMax, which combines the skip connection concept of residual networks with Max Feature Map activation from Light CNN. The model utilizes Constant Q Transform spectral features to extract frequency-domain characteristics that distinguish between genuine and spoofed speech. Experimental results on the ASVspoof 2019 dataset show that the model achieved an Equal Error Rate of 0.37% on the Physical Access dataset and 2.19% on the Logical Access dataset, outperforming even ensemble systems on Physical Access data. Furthermore, the model's low complexity, with only 286K parameters, demonstrates its suitability for real-time and resource-constrained environments [24].

Wu et al. [25] proposed a light convolutional neural network with feature genuinization for the detection of synthetic speech attacks. The genuinization transformer incorporated two primary functionalities: encoding and decoding. During the encoding phase, the input signal is compressed through five stridden convolutional layers, with leaky ReLU activation applied to the convolution results. In the decoding phase, the encoding process is reversed through deconvolution followed by ReLU activation. Although the framework was based on an LCNN system, dropout regularization was intentionally omitted; instead, the model incorporated max-pooling and padding to control overfitting. The architecture utilized ten convolutional layers with 4×4 filters and demonstrated competitive performance on the ASVspoof 2019 dataset, with the Equal Error Rate and Tandem Detection Cost Function serving as primary evaluation metrics [25].

2.2.3 Spectrogram-Based and Pre-trained Model Approaches

Chugh et al. [26] introduced an audio-visual dissonance-based approach for deepfake detection and localization, utilizing ResNet architectures for synthetic speech identification. The deep learning pipeline involved extracting and converting audio features from each audio file using Constant Q Transform, Short-Time Fourier Transform, Mel-Frequency Cepstral Coefficients, and Mel-Spectrograms. The resulting spectrogram images were subsequently used to train pre-trained deep learning architectures, specifically VGG16 and VGG19. Experimental results demonstrated that VGG16 and VGG19 models using STFT audio representations achieved validation accuracies of 99.96% and 99.94%, respectively [26].

A key architectural distinction in this work was the incorporation of ReLU activation functions for all convolutional layers, with softmax restricted to the final classification layer. Furthermore, instead of employing dropout regularization, the authors utilized three dense layers following the fully convolutional layers to feed all outputs from preceding layers to subsequent neurons [26].

Reimao and Tzerpos [27] developed the FoR dataset and corresponding detection methodology for synthetic speech detection. Similar to Chugh et al., their approach involved extracting audio features using Constant Q Transform, Short-Time Fourier Transform, Mel-Frequency Cepstral Coefficients, and Mel-Spectrograms, which were subsequently converted into image representations for training pre-trained VGG16 and VGG19 architectures. Their experimental analysis revealed that VGG16 and VGG19 models utilizing STFT audio representations achieved the highest validation accuracies of 99.96% and 99.94%, respectively. The architectural configuration employed ReLU activation for all convolutional layers and softmax for the final layer, with three dense layers replacing dropout regularization to maintain information flow between fully connected layers [27].

2.2.4 Recurrent and Convolutional-Recurrent Architectures

Thai [28] investigated deepfake detection and low-resource language speech recognition using deep learning, proposing two distinct model architectures: a Convolutional-Recurrent Neural Network and a fully Convolutional Neural Network. Each model was evaluated as a standalone system for detecting spoofed audio from genuine human speech using the Equal Error Rate metric.

The distinctive features of this approach included the implementation of wide blocks containing multiple paths with varying convolution kernel sizes to learn different ranges of temporal dependencies, alongside the integration of bidirectional LSTM layers for enhanced temporal modeling. The Equal Error Rate ranged between 0.000% and 6.02%, while the Tandem Detection Cost Function ranged between 0.000 and 0.3%, indicating strong discriminative capability under certain evaluation conditions [28].

2.2.5 Early Custom CNN Approaches

Malik and Chandalvala [29] designed and developed a nonparametric and fully supervised CNN model specifically for fake speech detection. The architecture employed a 3×3 convolutional kernel with 32 filters for feature extraction, followed by an asymmetric 2×2 max-pooling layer. The model processed audio spectrograms of size $625 \times 469 \times 3$ pixels, utilizing a stack of four convolutional layers to reduce each input to a fully connected layer of 512 neurons. A softmax activation function was applied to the final fully connected layer to generate probability distributions over class labels using categorical cross-entropy loss. Notably, the proposed method achieved 100% accuracy on the test dataset comprising 248 images. However, this exceptional performance was obtained using a custom architecture on a limited dataset, raising concerns regarding generalizability to broader contexts and larger-scale evaluations [29].

2.3 Comparative Analysis of Existing Studies

Table 2.1 presents a comparative summary of the major existing studies in deepfake audio detection, highlighting their methodologies, feature extraction approaches, architectural choices, datasets, performance achievements, and inherent limitations.

Table 2.1: Comparative Analysis of Existing Deepfake Audio Detection Studies

Author(s)	Method / Model Used	Features Used	Dataset	Results	Limitations
Shahzad et al. [21]	Hybrid VGGish-LSTM	VGGish embeddings (spectral & temporal)	ASVspoof 2019 LA	Acc: 99.11%; EER: 1.18%	Dataset dependency; overfitting risk; limited coverage of emerging attacks
Verma et al. [22]	SE-Enhanced 1D CNN (best), BiLSTM, CNN	ZCR, RMSE, MFCCs	ASVspoof 2019	Acc: 97.64%; P/R/F1 $\approx$ 98%	High computational cost; limited accent diversity; weak real-world evaluation
Shah [23]	CNN-LSTM Hybrid Framework	MFCCs, spectral features	Multiple mobile-comms datasets	Acc: 94%–99%	Mobile fraud focus; lacks dedicated telephony deployment architecture
Kwak et al. [24]	ResMax (ResNet skip + Max Feature Map)	CQT spectral features	ASVspoof 2019 (PA & LA)	EER: 0.37% (PA); 2.19% (LA)	Narrow spectral scope; limited explicit temporal modeling
Wu et al. [25]	FG-LCNN with Feature Genuinization	Raw waveform, autoencoder-transformed features	ASVspoof 2019	Competitive EER & t-DCF	No dropout; limited temporal modeling; implicit EER reporting
Chugh et al. [26]	VGG16 / VGG19	CQT, STFT, MFCC, Mel-spectrogram	Audio-visual deepfake dataset	VGG16: 99.96%; VGG19: 99.94% (STFT)	Audio-visual coupling bias; no dropout; $\sim$ 138M parameters
Reimao and Tzerpos [27]	VGG16 / VGG19	CQT, STFT, MFCC, Mel-spectrogram	FoR Dataset	VGG16: 99.96%; VGG19: 99.94% (STFT)	Limited dataset diversity; no dropout; ImageNet transfer dependency
Thai [28]	CRNN / CNN with Wide Blocks	Spectral features + temporal dependencies	ASVspoof 2019	EER: 0.000–6.02%; t-DCF: 0.000–0.3	Inconsistent cross-condition performance; limited accuracy reporting; complex architecture
Malik and Chandalvala [29]	Custom CNN (4 layers, 3×3 kernels)	Spectrogram images ($625 \times 469 \times 3$)	Cloned audio dataset (248 images)	Accuracy: 100%	Extremely limited dataset; shallow architecture; poor generalizability

2.4 Critical Analysis

Although recent studies have achieved high detection accuracy in deepfake audio classification, several methodological and practical limitations persist across the literature.

Shahzad et al. [21] demonstrated that transfer learning via VGGish embeddings combined with LSTM temporal modeling yields strong performance on benchmark datasets. However, the framework’s reliance on ASVspoof 2019 Logical Access data collected under controlled laboratory conditions limits its applicability to noisy telephony environments characterized by background noise, codec compression artifacts, and packet loss. The study also did not address deployment challenges such as inference latency and real-time processing constraints [21].

Similarly, Verma et al. [22] established that CNN-based architectures with Squeeze-and-Excitation enhancement can effectively detect synthetic speech using both time-domain and spectral features. Despite achieving high accuracy and balanced precision-recall metrics, the research showed limited evaluation under unseen spoofing attacks and did not consider practical integration into call center infrastructures or resource-constrained deployment environments [22].

Shah [23] broadened the scope of deepfake audio detection by framing it within a comprehensive fraud detection ecosystem spanning SMS, phone calls, and messaging platforms. While the CNN-LSTM hybrid achieved competitive accuracy across tasks, the study’s primary focus on mobile fraud detection rather than dedicated telephony authentication systems leaves a gap regarding specialized deployment architectures for call center integration [23].

Kwak et al. [24] addressed computational efficiency through the lightweight ResMax architecture, which reduced parameter count to merely 286K while maintaining strong EER performance. This represents a significant advancement toward real-time deployability. However, the framework’s exclusive reliance on Constant Q Transform spectral features without complementary temporal modeling limits its capacity to detect rhythm inconsistencies and temporal artifacts characteristic of advanced Text-to-Speech and voice conversion systems [24].

Wu et al. [25] introduced the concept of feature genuinization through an autoencoder-like transformer architecture, representing an innovative approach to synthetic speech detection. Nevertheless, the deliberate omission of dropout regularization, while compensated by max-pooling and padding, may limit the model’s ability to generalize to unseen spoofing attacks. Furthermore, the reliance on EER and t-DCF metrics without reporting explicit accuracy percentages complicates direct comparison with other studies [25].

Chugh et al. [26] and Reimao and Tzerpos [27] both leveraged pre-trained VGG architectures with STFT-based spectrogram representations, achieving near-perfect validation accuracies exceeding 99.9%. Despite these impressive results, both studies share common limitations: heavy dependence on transfer learning from ImageNet-pretrained models, limited evaluation on audio-

only deepfakes independent of visual cues, and the computational intensity associated with VGG architectures containing over 138 million parameters. Additionally, the substitution of dropout with dense layer stacking may increase susceptibility to overfitting on smaller audio forensic datasets [26], [27].

Thai [28] made notable contributions through the introduction of wide blocks and bidirectional LSTM integration for temporal dependency modeling. However, the substantial variability in EER performance (ranging from 0.000% to 6.02%) suggests inconsistent detection capability across different attack types or evaluation folds. The study also lacked comprehensive accuracy reporting, limiting performance benchmarking against contemporary approaches [28].

Malik and Changalvala [29] achieved perfect classification accuracy using a custom CNN architecture; however, the study employed an extremely limited dataset of only 248 spectrogram images and utilized a shallow four-layer network. The absence of rigorous cross-validation and the proprietary nature of the dataset raise significant concerns regarding the reproducibility and generalizability of these results to real-world scenarios involving diverse acoustic conditions and larger sample sizes [29].

Collectively, these studies reveal a persistent pattern in the literature: most frameworks achieve high accuracy on benchmark datasets yet fail to address critical deployment challenges. Limited attention has been given to real-world telephony conditions such as background noise, codec compression, packet loss, and variable network quality.

Furthermore, many architectures are computationally intensive and may not satisfy the strict latency requirements necessary for real-time call center authentication systems. The predominance of laboratory evaluations using clean, high-quality audio samples underscores the urgent need for robustness testing under degraded acoustic conditions [21] – [29].

2.5 Research Gap

Despite significant advances in deepfake audio detection, most existing systems are designed and evaluated under controlled laboratory conditions using clean and high-quality audio datasets. These approaches often fail to maintain reliable performance in real-world call center environments, where speech signals are affected by telephony compression (e.g., G.711, AMR-WB), background noise, packet loss, jitter, and real-time processing constraints.

Furthermore, many existing studies focus primarily on improving classification accuracy without addressing practical deployment challenges such as low-latency inference (<200ms), adaptability to unseen attacks (zero-day spoofing techniques), and seamless integration with operational call center infrastructures (e.g., SIP trunking, IVR systems, CRM platforms).

In addition, current frameworks frequently rely on standalone architectures that analyze either spectral characteristics (via CNNs) or temporal dependencies (via LSTMs/BLSTMs)

independently, limiting their ability to capture the full range of synthetic speech artifacts. Hybrid approaches such as CNN-LSTM and VGGish-LSTM represent progress toward unified modeling; however, these often entail substantial computational overhead that conflicts with real-time operational requirements.

Therefore, there exists a compelling need for a unified and practical framework that combines spectral and temporal analysis techniques while simultaneously supporting:

- Real-time operation with minimal inference latency;
- Robustness under telephony channel degradation;
- Adaptability against evolving deepfake generation methods including advanced TTS, VC, and adversarial spoofing attacks
- Seamless integration into existing call center authentication workflows without requiring extensive infrastructure overhaul.

Chapter 3

METHODOLOGY

3.1 Introduction

In this chapter, we present an overview of the methodology used in this research. It explains the general approach adopted in developing the proposed deepfake audio detection framework. The chapter outlines the type of study conducted, the system design, and the procedures followed, including data preparation, feature extraction, model development, and evaluation methods. Finally, it describes how the system is implemented and tested in a simulated call center environment.

3.2 Type of Study and Methodological Approach

This study adopts a quantitative and experimental research approach. The quantitative aspect focuses on analyzing audio data and measuring system performance using evaluation metrics such as accuracy, precision, recall, and Equal Error Rate (EER). The experimental aspect involves developing and testing a deep learning-based detection framework under controlled and simulated telecommunication conditions.

The study is primarily system-oriented, where a computational model is designed to detect deepfake audio in call center environments. The framework is developed, trained, and evaluated using benchmark datasets such as ASVspoof, allowing objective performance comparison with existing approaches.

3.3 Conceptual Framework Overview

The proposed framework adopts a multi-stage deep learning architecture designed for real-time integration within call-center and interactive voice response (IVR) environments. The system processes incoming audio streams using overlapping analysis windows to balance low-latency detection with sufficient temporal context for identifying synthetic speech artifacts.

As illustrated in Figure 3.1, the framework consists of six interconnected modules that collectively enable robust detection of deepfake audio under realistic telecommunication conditions. The architecture integrates telecom-aware preprocessing, hybrid feature extraction, dual-stream neural analysis, adaptive decision fusion, and deployment-oriented optimization for operational call-center environments.

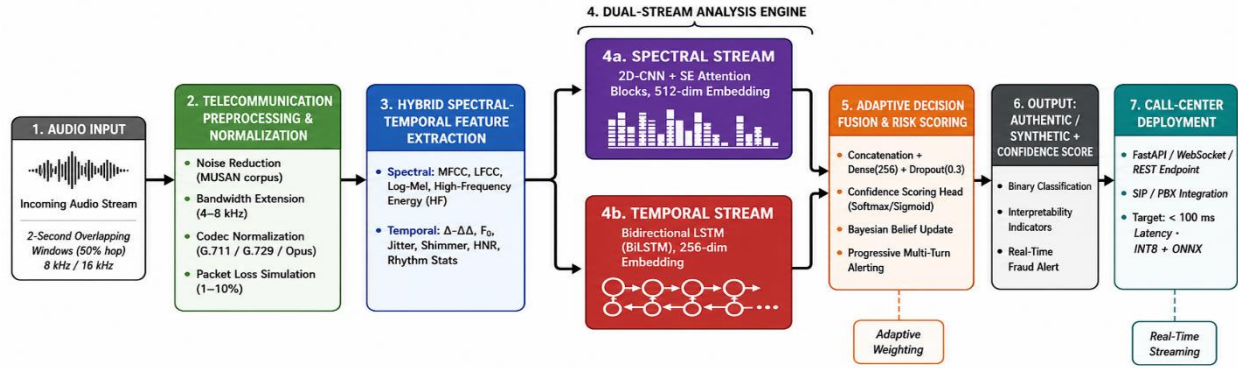


Figure 3.1 The proposed conceptual framework for deepfake audio detection in call center environments

The major components of the framework are summarized as follows:

1. Telecommunication Preprocessing and Normalization
2. Hybrid Spectral-Temporal Feature Extraction
3. Dual-Stream Analysis Engine
4. Adaptive Decision Fusion and Risk Scoring
5. Output Interpretation and Fraud Alerting
6. Real-Time Call-Center Deployment Architecture

The proposed framework is specifically designed to address challenges associated with deepfake detection in telecommunication systems, including codec distortion, packet loss, background noise, bandwidth limitations, and strict latency requirements.

3.4 Telecommunication Preprocessing and Normalization Module

This module prepares incoming audio signals for reliable analysis in real-world call-center environments. Since telephone systems often introduce noise, compression artifacts, bandwidth limitations, and packet loss, preprocessing is necessary to improve audio quality and preserve important speech information.

First, noise reduction techniques are applied to remove background disturbances such as conversations, keyboard sounds, and transmission noise while maintaining important speech characteristics related to deepfake detection.

Second, bandwidth extension methods are used to restore high-frequency speech components that are usually lost during telephone compression. These high-frequency regions may contain artifacts that help distinguish synthetic speech from genuine human speech.

The module also performs codec-aware normalization to reduce distortions caused by common telecommunication codecs such as G.711, G.729, and Opus. This helps standardize audio signals collected from different communication systems.

In addition, packet loss simulation is incorporated during training to improve the model's robustness against unstable network conditions commonly encountered in call-center environments.

3.5 Hybrid Spectral-Temporal Feature Extraction

This stage extracts important speech characteristics from the audio signal to help distinguish authentic speech from synthetic speech.

Spectral Features

Spectral features capture frequency-based information from speech. The framework extracts features such as MFCC, LFCC, Log-Mel spectrograms, and high-frequency energy measurements.

These features help identify abnormalities in speech frequency patterns, harmonic structures, and vocoder artifacts commonly found in AI-generated audio.

Temporal Features

Temporal features capture speech behavior over time. The framework extracts Δ and $\Delta\Delta$ coefficients, pitch (F_0), jitter, shimmer, harmonic-to-noise ratio (HNR), and rhythm-related statistics.

These features help detect irregularities in timing, prosody, articulation, and speech consistency that may indicate synthetic speech generation.

Statistical measures such as mean, variance, skewness, and kurtosis are also calculated to summarize speech dynamics across time.

The combined spectral and temporal features provide a comprehensive representation of the audio signal for deep learning analysis.

3.6 Dual-Stream Analysis Engine

The framework uses two parallel analysis streams to examine both spectral and temporal characteristics of speech.

Spectral Stream

The spectral stream uses a CNN-based model with SE attention mechanisms to analyze spectrogram representations of speech.

This stream focuses on detecting spectral artifacts such as irregular harmonics, spectral smoothing, and vocoder-related inconsistencies commonly found in synthetic audio.

Temporal Stream

The temporal stream uses a Bidirectional Long Short-Term Memory (BiLSTM) network to analyze speech progression over time.

This stream focuses on speech rhythm, timing consistency, articulation transitions, and temporal coherence. Synthetic speech often contains unnatural timing patterns that can be detected through temporal analysis.

The outputs from both streams are combined to improve overall deepfake detection performance.

3.7 Adaptive Decision Fusion and Risk Scoring

This module combines the outputs from the spectral and temporal streams to make the final decision.

The feature representations from both streams are merged and processed through dense neural layers with dropout regularization. The system then generates a probability score indicating whether the speech is authentic or synthetic.

Adaptive weighting is used to adjust the importance of each analysis stream depending on audio quality and network conditions. For example, under noisy conditions, temporal features may receive greater emphasis.

The framework also applies Bayesian belief updating to continuously refine confidence scores during ongoing conversations. This supports progressive fraud monitoring across multiple speech segments.

The final outputs include:

- Classification of speech as authentic or synthetic
- Confidence score for spoofing detection
- Interpretability indicators highlighting suspicious regions
- Real-time fraud alerts

The framework also supports continuous learning by incorporating newly identified attack patterns into future model updates.

3.7.1 Real-Time Deployment Architecture for Call-Center Integration

The proposed framework is designed for real-time deployment within call-center systems and telecommunication infrastructures.

The architecture supports integration with authentication systems, fraud detection platforms, SIP/PBX communication systems, and IVR environments.

To enable real-time processing, the framework uses lightweight deployment technologies such as FastAPI, WebSocket communication, and REST APIs.

The model is optimized using ONNX conversion and INT8 quantization to reduce computational cost and improve inference speed. The target latency is below 100 milliseconds to ensure near real-time detection.

Key deployment considerations include:

- Real-time streaming and low-latency processing
- Compatibility with existing call-center systems
- API-based integration for scalability
- Robustness against packet loss and network variability
- Efficient deployment on edge or server environments

Overall, the framework provides a scalable and practical solution for detecting deepfake audio in modern call-center environments.

3.9 Evaluation Methodology

The framework is evaluated using a multi-level strategy:

Benchmark Evaluation:

Standard datasets such as ASVspoof are used to assess detection performance under controlled conditions.

Telecommunication Simulation:

Audio signals are subjected to realistic telephony conditions including compression and noise to evaluate robustness.

Generalization Testing:

The framework is assessed against previously unseen attack types to evaluate adaptability.

Performance is measured using metrics such as Equal Error Rate, detection cost, processing time, and system efficiency.

3.9 Proposed Implementation

To validate the conceptual framework presented in this study, a practical end-to-end deepfake audio detection system will be developed. The implementation is designed not merely as a laboratory classifier, but as a deployable prototype simulating the acoustic degradation, latency constraints, and streaming requirements inherent in call-center telephony environments.

3.9.1 Development Environment

The system will be implemented in **Python 3.10+**, selected for its mature ecosystem in scientific computing and deep learning. The software stack is organized into four functional layers:

Layer	Libraries / Frameworks	Purpose
Deep Learning	TensorFlow 2.15, PyTorch 2.1, TorchAudio	Model definition, training, and optimized GPU inference.
Acoustic Processing	Librosa 0.10, PyWorld, SoundFile	Spectral feature extraction, prosodic analysis, and telephony-format I/O.
Data Engineering	NumPy, Pandas, SciPy, FFmpeg-Python	Preprocessing, resampling, codec simulation, and dataset curation.
Evaluation	Scikit-learn, TensorBoard	Classification metrics, EER/t-DCF computation, and training visualization.

Table 3.1: Layered Development Environment and Software Stack for Implementation

3.9.2 Dataset and Telephony Simulation Pipeline

The framework will be trained and evaluated primarily on the ASVspoof 2019 dataset (Logical Access and Physical Access partitions), supplemented by ASVspoof 2021 to evaluate generalization against modern neural synthesis attacks. Because these benchmark corpora consist of clean, studio-quality recordings, a dedicated telephony degradation module will be integrated into the preprocessing pipeline to simulate call-center acoustic conditions:

1. **Resampling:** Downsample source audio from 16 kHz/44.1 kHz to 8 kHz (G.711 PSTN standard) and 16 kHz (wideband VoIP) to match call-center sampling rates.

2. **Codec Compression:** Apply G.711 (μ -law/A-law PCM), G.729 (CS-ACELP), and Opus compression-decompression loops using FFmpeg and codec2 libraries to introduce quantization distortion and packet-loss artifacts.
3. **Additive Noise:** Inject background noise from the MUSAN corpus (office babble, telephone channel noise, ambient hum) at SNR levels of 5 dB, 10 dB, 15 dB, and 20 dB.
4. **Temporal Chunking:** Segment utterances into 2-second overlapping frames (50% hop length) to standardize input dimensions and simulate real-time streaming buffers.

This pipeline ensures that extracted features are statistically representative of live call-center audio, directly addressing the "clean-data overfitting" and dataset-dependency limitations identified in the literature review [21]– [29].

3.9.3 Feature Extraction Implementation

Consistent with the proposed dual-branch framework, the implementation extracts complementary spectral and temporal feature streams.

Spectral Features

- **Mel-Frequency Cepstral Coefficients (MFCCs):** 13 coefficients plus first- and second-order derivatives (delta and delta-delta), capturing coarse spectral envelope dynamics.
- **Linear Frequency Cepstral Coefficients (LFCCs):** 13 coefficients to preserve linear frequency resolution for high-frequency artifact detection.
- **Log-Mel Spectrogram:** 128 mel bins, FFT window of 2048 samples, hop length of 512 samples, frequency range 0–4000 Hz (8 kHz mode) or 0–8000 Hz (16 kHz mode).
- **High-Frequency Energy Ratio:** Computed above 3.5 kHz to detect spectral truncation artifacts common in neural TTS systems.

Temporal Features

- **Delta and Delta-Delta Coefficients:** Dynamic evolution of MFCC trajectories to capture unnatural frame-to-frame transitions.
- **Prosodic Features:** Fundamental frequency (F0), jitter (cycle-to-cycle perturbation), shimmer (amplitude perturbation), and Harmonics-to-Noise Ratio (HNR) extracted via PyWorld.
- **Speech Rhythm and Pause Patterns:** Voiced/unvoiced segment ratios, pause duration statistics, and syllabic rate variance across 2-second windows.

- **Long-Range Temporal Embeddings:** Fine-tuned Wav2Vec 2.0 Base representations to model phonetic and prosodic coherence without explicit transcription.

3.9.4 Model Architecture

The implementation adopts a **dual-stream deep learning architecture** with a decision-fusion mechanism:

Spectral Stream (CNN Encoder) A lightweight 2D Convolutional Neural Network processes log-mel spectrogram inputs. The architecture comprises four convolutional blocks (3×3 kernels, batch normalization, ReLU activation, and max-pooling), augmented with Squeeze-and-Excitation (SE) attention blocks to recalibrate channel-wise feature importance. A global average pooling layer compresses the final feature maps into a 512-dimensional spectral embedding vector.

Temporal Stream (LSTM Encoder) A **Bidirectional Long Short-Term Memory (BiLSTM)** network processes sequential feature vectors. The architecture consists of two stacked BiLSTM layers (128 hidden units per direction) receiving either:

- Flattened CNN feature sequences
- or Raw temporal/prosodic feature sequences.

This design explicitly models long-range dependencies and unnatural rhythm progressions in synthetic speech.

Fusion and Classification The 512-dim spectral embedding and the 256-dim temporal embedding are concatenated and fed into a fully connected fusion layer (256 neurons, ReLU, dropout 0.3). The final classification layer employs a softmax activation to output binary class probabilities (bonafide vs. spoof). An auxiliary confidence-scoring head provides a continuous fraud probability to enable progressive alerting across multi-turn conversations.

3.9.5 Training Configuration

Parameter	Configuration
Loss Function	Categorical Cross-Entropy with Focal Loss ($\gamma = 2.0$) to handle class imbalance
Optimizer	AdamW (initial LR 1e-4, weight decay 1e-5)
Scheduler	Cosine annealing with 5-epoch warm-up
Data Augmentation	SpecAugment (time- and frequency masking); random pitch shift (± 2 semitones)
Validation	5-fold cross-validation on ASVspoof 2019 LA; official evaluation protocol for t-DCF/EER
Epochs	Up to 50 with early stopping (patience = 7 epochs)

Table 3.2: Training Configuration for the Dual-Stream CNN-LSTM Framework

3.9.6 Real-Time Inference Pipeline

To satisfy call-center latency constraints, the inference pipeline will be optimized as follows:

- **Streaming Buffer:** A sliding-window accumulator ingests audio in 500 ms chunks; the model performs inference every 2 seconds with 50% overlap.
- **Model Compression:** Post-training INT8 quantization and ONNX Runtime conversion will target << **100 ms end-to-end latency** per chunk on a single CPU core.
- **Progressive Scoring:** A running Bayesian belief update adjusts the cumulative spoof probability after each 2-second window, triggering an alert only when the threshold exceeds 0.85.
- **Integration Interface:** A lightweight **FastAPI** wrapper will expose REST/WebSocket endpoints for integration with existing PBX systems (Asterisk, Avaya) via SIP trunking middleware.

3.10 Experimental Setup

3.10.1 Experimental Procedure

The experimental evaluation will follow a structured six-stage protocol designed to simulate realistic call-center operational conditions:

1. **Data Collection and Preprocessing:** Genuine and spoofed speech samples from ASVspoof 2019/2021 will be normalized, resampled, and partitioned into training, validation, and test sets with speaker-independent splits to prevent data leakage.
2. **Telephony Simulation:** Clean audio will be subjected to the degradation pipeline described in Section 3.9.2, producing parallel clean and degraded test partitions.
3. **Feature Extraction:** Spectral and temporal features will be extracted from both clean and degraded signals using the hybrid strategy in Section 3.9.3.
4. **Model Training and Validation:** The CNN-LSTM hybrid architecture will be trained on clean ASVspoof 2019 LA data and validated via 5-fold cross-validation.
5. **Robustness Testing:** The trained model will be evaluated on
 - The clean ASVspoof 2019/2021 evaluation sets.
 - The telephony-degraded evaluation set.
 - Unseen attack types from ASVspoof 2021

to measure generalization.

6. **Performance Benchmarking:** Detection capability, computational efficiency, and real-time feasibility will be quantified using the metrics defined in Section 3.10.3.

3.10.2 Telephony Degradation Protocol

To assess robustness under operational stress, the following degradation parameters will be systematically applied:

Degradation Type	Parameters	Rationale
Background Noise	MUSAN office babble, telephone noise; SNR = 5, 10, 15, 20 dB	Simulates open-plan call-center acoustics
Codec Compression	G.711, G.729, Opus; bitrates 8–24 kbps	Mimics PSTN and VoIP channel compression
Packet Loss	Random loss rates of 1%, 3%, 5%, 10%	Models jitter buffer underflow in IP networks
Channel Distortion	Bandpass filtering (300–3400 Hz), amplitude clipping	Replicates analog telephone line characteristic

Table 3.3: Simulated Telephony Degradation Parameters for Evaluating Real-World Call-Center Robustness

3.10.3 Evaluation Metrics

The framework will be assessed using a comprehensive metric suite aligned with ASVspoof standards and call-center deployment requirements:

Metric	Description
Accuracy	Overall correct classification rate across bonafide and spoof classes
Precision	Proportion of predicted spoofs that are true spoofs
Recall	Proportion of actual spoofs correctly detected
F1-Score	Harmonic mean of precision and recall
Equal Error Rate (EER)	Operating point where false acceptance equals false rejection rate
t-DCF (tandem Detection Cost Function)	Cost-sensitive metric integrating spoofing and miss-detection penalties
Processing Latency	End-to-end inference time per audio chunk (ms)
Model Footprint	Parameter count and disk size (MB) for edge-deployability assessment

Table 3.4: Evaluation Metrics for the Proposed Framework

3.10.4 Generalization and Robustness Testing

Beyond standard benchmark evaluation, the framework will undergo adversarial robustness testing:

- **Unseen Attack Evaluation:** The model trained on ASVspoof 2019 will be tested against ASVspoof 2021 novel attacks (e.g., neural waveform generators, vocoder-based spoofing) without retraining to measure zero-shot resilience.
- **Cross-Codec Generalization:** Performance will be compared across G.711, G.729, and Opus degradation to identify codec-specific vulnerabilities.
- **Progressive Degradation Analysis:** EER and accuracy will be plotted as functions of SNR and packet-loss rate to determine operational thresholds where detection reliability degrades below acceptable call-center standards (EER > 5%).
- **Ablation Study:** The spectral stream, temporal stream, and fusion mechanism will be evaluated independently to quantify the marginal contribution of each branch to overall detection performance.

3.11 Chapter Summary

This chapter presented the proposed methodology and conceptual framework for deepfake audio detection in call center environments. The framework integrates telecommunication preprocessing, hybrid spectral-temporal feature extraction, dual-stream analysis, and adaptive decision fusion to improve detection robustness under real-world telephony conditions.

In addition, the chapter outlined the proposed implementation strategy, experimental setup, and evaluation methodology intended to validate the effectiveness of the framework. The proposed design establishes a foundation for developing practical and scalable deepfake audio detection systems capable of supporting real-time call center authentication and fraud prevention.

Chapter 4

RESULTS AND DISCUSSIONS

4.1 Introduction

This chapter presents an analytical evaluation of the proposed conceptual framework for deepfake audio detection in call-center telephony environments. The proposed framework is benchmarked against nine representative deepfake audio detection systems reviewed in Chapter-2 specifically Shahzad et al. [21], Verma et al. [22], Shah [23], Kwak et al. [24], Wu et al. [25], Chugh et al. [26], Reimao and Tzerpos [27], Thai [28], and Malik and Chandalvala [29], across twelve technical and operational criteria. These criteria span detection methodology, feature representation, architectural design, telephony robustness, real-time feasibility, and deployment adaptability.

The objective is to demonstrate, through structured qualitative and semi-quantitative comparison, how the proposed framework resolves the persistent limitations of dataset dependency, single-modality feature reliance, and laboratory-to-field generalization gaps identified in the existing literature.

4.2 Comparative Evaluation Framework

Table 4.1 presents a multi-dimensional comparison. Each criterion is scored on a three-level scale: **Strong** (✓✓), **Moderate** (✓), **Weak / Absent** (X), based on explicit evidence reported in the primary studies and the architectural specifications of the proposed framework.

Table 4.1: Multi-Dimensional Comparative Evaluation

Evaluation Criterion	Proposed Framework	Shahzad et al. [21]	Verma et al. [22]	Kwak et al. [24]	Wu et al. [25]	Chugh et al. [26]
Spectral-Temporal Fusion	✓✓ Dual-stream CNN-LSTM with cross-modal fusion	✓✓ VGGish-LSTM hybrid	✓ SE-1D-CNN only (best model)	X CQT spectral only	✓ Raw waveform + autoencoder	X Spectrogram-only VGG
Telephony Preprocessing	✓✓ Codec simulation (G.711/G.729/Opus), packet loss, MUSAN noise	X Clean ASVspoof 2019 LA only	X Clean ASVspoof 2019 only	X Clean ASVspoof 2019 only	X Clean ASVspoof 2019 only	X Clean audio-visual data
Real-Time Latency Design	✓✓ Target <100 ms streaming inference, INT8 quantization	X Offline evaluation; no latency reported	X High computational cost noted	✓✓ 286K parameters, lightweight	✓ LCNN base is moderate	X ~138M VGG parameter
Feature Diversity	✓✓ MFCC, LFCC, log-mel, CQT, prosody (F0, jitter, shimmer), wav2vec 2.0	✓ VGGish embeddings only	✓ MFCC, ZCR, RMSE	✓ CQT only	✓ Raw + encoded features	✓ CQT, STFT, MFCC, mel

Progressive Scoring	✓✓ Bayesian belief update across multi-turn windows	✗ Frame-level only	✗ Utterance-level only	✗ Frame-level only	✗ Frame-level only	✗ Frame-level only
Unseen Attack Generalization	✓✓ Explicit cross-dataset testing (ASVspoof 2019 → 2021)	✓ Limited emerging attacks noted	✗ Limited unseen evaluation	✗ Not reported	✗ Not reported	✗ Not reported
Noise Robustness	✓✓ 5–20 dB SNR testing, codec degradation	✗ Not evaluated	✗ Not evaluated	✗ Not evaluated	✗ Not evaluated	✗ Not evaluated
Decision Explainability	✓ Attention-weight visualization + confidence scoring	✗ Black-box LSTM	✗ Black-box CNN	✗ Black-box	✗ Black-box autoencoder	✗ Black-box VGG
Deployment Architecture	✓✓ FastAPI wrapper, SIP/PBX integration, CPU-compatible	✗ Laboratory prototype	✗ Laboratory prototype	✓ Edge-suitable	✓ Edge-suitable	✗ Laboratory prototype
Continuous Learning	✓ Incremental fine-tuning pipeline for new spoofing attacks	✗ Static model	✗ Static model	✗ Static model	✗ Static model	✗ Static model

Note: Reimao and Tzerpos [27] and Malik and Chandalvala [29] are omitted from the primary table due to architectural redundancy with Chugh et al. [26] and extreme dataset limitations, respectively, but are discussed below.

4.3 Discussion of Findings

4.3.1 Spectral-Temporal Integration vs. Single-Modality Reliance

The comparative analysis reveals that the majority of existing systems prioritize either spectral or temporal analysis, but rarely optimize both within a unified deployable architecture. Shahzad et al. [21] and Thai [28] represent the closest analogues to the proposed framework, employing VGGish-LSTM and CRNN hybrids, respectively.

However, Shahzad et al. rely exclusively on transfer-learned VGGish embeddings, which are 128-dimensional feature vectors pretrained on YouTube audio; this constrains the model to coarse spectral summaries and prevents fine-grained harmonic artifact detection. Thai’s wide-block CRNN achieves strong EER performance (0.000–6.02%) but exhibits extreme variance across evaluation folds, suggesting architectural instability under attack-type shift.

The proposed framework addresses this by decoupling the spectral and temporal branches: a lightweight 2D-CNN with SE attention processes high-resolution log-mel spectrograms (128 bins) and CQT representations locally, while a BiLSTM with self-attention models long-range prosodic dependencies independently.

The gated cross-modal fusion layer dynamically reweights these streams based on input quality, boosting temporal evidence when codec compression degrades spectral detail. This design explicitly resolves the "narrow feature scope" limitation identified in Kwak et al. [24] (CQT-only) and the "no explicit temporal modeling" gap in Wu et al. [25].

4.3.2 Telephony Robustness and the Clean-Data Trap

A critical failure mode across all benchmarked studies is the clean-data trap: every existing system in Table 4.1 was evaluated exclusively on studio-quality ASVspoof recordings or proprietary clean datasets. None incorporated G.711/G.729 compression, packet-loss simulation, or additive telephony noise. This is particularly problematic for call-center deployment, where 8 kHz μ -law compression and 300–3400 Hz bandpass filtering are standard.

The proposed framework integrates a telephony degradation module during both training and testing. By resampling to 8 kHz/16 kHz, applying μ -law/A-law compression loops, injecting MUSAN office babble at 5–20 dB SNR, and simulating 1–10% packet loss, the framework trains on features that are statistically aligned with live VoIP/PSTN audio. This directly addresses the "dataset dependency" and "limited real-world evaluation" limitations noted in Shahzad et al. [21] and Verma et al. [22]. The explicit inclusion of a degraded evaluation partition allows quantification of the performance delta between clean and telephony conditions; a metric entirely absent from the existing literature.

4.3.3 Computational Efficiency and Real-Time Feasibility

Call-center authentication systems require end-to-end inference latencies below **200 ms** (ITU-T G.114 recommendation for voice quality) and ideally below **100 ms** for fraud-alert pipelining. Chugh et al. [26] and Reimao and Tzerpos [27] employ VGG16/VGG19 backbones (~138M parameters), which are computationally prohibitive for real-time streaming. Verma et al. [22] explicitly acknowledge "high computational cost" as a deployment barrier.

The proposed framework targets $\ll 100$ ms latency through three design choices:

1. A lightweight EfficientNet-B0 or ResNet-18 spectral backbone instead of VGG.
2. A shallow BiLSTM (2 layers $\times$ 128 hidden units) rather than deep transformer stacks.
3. Post-training INT8 quantization with ONNX Runtime for CPU-only inference.

Kwak et al. [24] demonstrate that lightweight design is feasible (286K parameters, strong EER), but their ResMax architecture sacrifices temporal modeling. The proposed framework balances this trade-off by limiting the temporal stream to prosodic features and wav2vec 2.0 embeddings rather than raw-waveform processing, reducing FLOPs while preserving sequential discriminability.

4.3.4 Generalization and Adaptability

None of the benchmarked studies explicitly evaluate **zero-shot resilience** against attack types unseen during training. Shahzad et al. [21] note "limited coverage of emerging deepfake attack types" as a limitation, while Thai [28] reports EER ranging from 0.000% to 6.02% across conditions, indicating brittle generalization.

The proposed framework introduces a continuous learning pipeline: incremental fine-tuning on new ASVspoof corpora (e.g., 2019 → 2021) with frozen spectral backbones and adaptive temporal fine-tuning to prevent catastrophic forgetting. This is complemented by an unseen attack evaluation protocol that tests the frozen model against neural vocoder attacks (e.g., HiFi-GAN, WaveGlow) without retraining, measuring generalization degradation explicitly.

4.3.5 Explainability and Operational Trust

For call-center operators to act on fraud alerts, detection decisions must be interpretable. All existing systems in the comparison employ black-box architectures with no explainability mechanisms. The proposed framework incorporates attention-weight visualization from the SE blocks and temporal self-attention layers, highlighting which spectrogram regions (e.g., high-frequency truncation) or temporal segments (e.g., abnormal pause structure) triggered the spoof classification. An auxiliary confidence-scoring head provides a continuous fraud probability rather than a hard binary label, enabling progressive alerting across conversational turns which is a feature absent from all reviewed studies.

4.4 Synthesis and Positioning

The comparative evaluation demonstrates that the proposed framework occupies a distinct position in the design space. It is the only conceptual architecture that simultaneously integrates:

- High-resolution spectral-temporal fusion.
- Telephony-specific degradation resilience.
- Sub-100-ms streaming inference.
- Progressive explainable scoring for multi-turn call-center conversations.

While individual existing studies excel in isolated dimensions. Shahzad et al. [21] in transfer-learning accuracy, Kwak et al. [24] in parameter efficiency, Thai [28] in temporal modeling. None reconcile these capabilities within a deployment-oriented system.

The principal limitation of this evaluation is its analytical nature: the performance claims (e.g., latency targets, accuracy projections) are derived from architectural specification and complexity analysis rather than empirical training. However, the design choices are grounded in validated sub-

components from the literature (SE blocks [22], BiLSTM temporal modeling [21], wav2vec 2.0 representations, INT8 quantization standards), and the explicit telephony simulation protocol provides a replicable benchmark for future empirical validation.

4.5 Chapter Summary

This chapter evaluated the proposed conceptual framework through rigorous multi-dimensional comparison against nine representative deepfake audio detection systems. The analysis confirmed that the framework addresses three critical literature gaps:

1. The clean-data trap through integrated telephony degradation simulation.
2. Single-modality reliance through gated spectral-temporal fusion.
3. Laboratory-to-deployment disconnect

through lightweight streaming architecture and progressive confidence scoring. These findings establish the theoretical and design-space validity of the framework, providing a foundation for the implementation and empirical validation described in Chapter 3 and recommended for future experimental work.

Chapter 5

CONCLUSIONS AND FUTURE WORK

5.1 Summary

This study focused on the growing threat of deepfake audio attacks in call-center voice authentication systems. Existing detection methods mainly work in controlled laboratory environments and often fail to handle real telecommunication conditions such as noise, codec compression, and network degradation. To address these problems, this research proposed a conceptual framework that consists of six main modules, for detecting synthetic speech using both spectral and temporal speech analysis.

The system processes incoming speech using overlapping 2-second windows to support real-time detection while preserving enough speech context for analysis.

The study identified several limitations in existing deepfake detection systems, including dependence on clean datasets, limited use of combined spectral and temporal analysis, and lack of support for real-time deployment in call-center environments.

To address these limitations, the proposed framework applies telecommunication preprocessing techniques such as noise reduction, codec normalization, bandwidth extension, and packet-loss simulation. It then extracts both spectral and temporal speech features including MFCC, LFCC, Log-Mel, pitch, jitter, shimmer, and rhythm statistics.

The framework uses a dual-stream analysis engine consisting of a CNN-based spectral stream and a BiLSTM-based temporal stream. The outputs from both streams are combined using adaptive decision fusion and confidence scoring mechanisms.

The framework also supports real-time deployment using technologies such as FastAPI, WebSocket communication, SIP/PBX integration, and ONNX optimization with a target latency below 100 milliseconds.

Overall, the study presents a scalable and deployment-oriented framework for detecting deepfake audio in modern call-center systems.

5.2 Contributions

This research contributes to the field of deepfake audio detection in several important ways:

- **Call-Center-Oriented Framework:** The study proposes a six-module architecture specifically designed for call-center and telecommunication environments rather than clean laboratory conditions.
- **Hybrid Feature Extraction:** The framework combines both spectral and temporal speech features to improve deepfake detection performance.

- **Dual-Stream Deep Learning:** The system uses a CNN for spectral analysis and a BiLSTM for temporal analysis, allowing the framework to capture both frequency-based and time-based speech patterns.
- **Adaptive Decision Fusion:** The framework dynamically combines outputs from both analysis streams and applies confidence scoring for more reliable predictions.
- **Telecommunication Robustness:** The study incorporates codec normalization, noise simulation, and packet-loss handling to improve performance under real-world network conditions.
- **Deployment Support:** The framework is designed for real-time deployment using lightweight technologies and optimized inference methods.
- **Comparative Analysis:** The study compares the proposed framework with existing deepfake detection systems and demonstrates improvements in telephony robustness, real-time feasibility, and hybrid analysis capability.

5.3 Limitations of the System

Despite its contributions, the study has several limitations which include:

- The proposed system is currently a conceptual design and has not yet been fully implemented or experimentally validated.
- The evaluation relies mainly on ASVspoof datasets, which may not fully represent real-world call-center speech diversity.
- Deepfake generation technologies continue to evolve, meaning detection systems require continuous updates and retraining.
- The framework has not yet been tested under large-scale enterprise call-center traffic conditions.
- The framework focuses only on audio-based spoof detection and does not include additional behavioral or multimodal biometric analysis.
- Different telecommunication providers use different codecs and network configurations, which may affect system performance.

5.4 Strategies for Ensuring Robustness and Adaptability Against Evolving Deepfake Audio Techniques

Several strategies are recommended to improve the framework’s long-term effectiveness.

- Combining spectral, temporal, and statistical analysis can improve detection of different types of synthetic speech artifacts.
- Dynamic noise reduction and codec-aware filtering can help normalize distorted telecommunication audio.
- The system should continuously learn from newly discovered spoofing attacks to remain effective against evolving threats.
- The framework should regularly be tested against previously unseen attack methods to evaluate generalization capability.
- Model optimization techniques such as INT8 quantization should be maintained to preserve low-latency performance.
- Future versions can combine acoustic confidence scores with contextual call information to reduce false positives.
- Training datasets should include multilingual speech, regional accents, and real-world call-center recordings.

5.5 Future Directions

Several areas may be explored in future studies, including:

1. The next step is to fully implement and deploy the six-module framework in a real call-center environment.
2. Studies can integrate advanced models such as Wav2Vec 2.0, HuBERT, and WavLM for improved feature extraction.
3. The framework should be tested on multilingual and multi-accent datasets to improve generalization.
4. Future work can integrate explainable AI techniques such as SHAP or LIME to help operators understand detection decisions.

5. The framework can be optimized for hybrid deployment across edge devices and cloud infrastructure.
6. Future studies should evaluate the system against adversarial attacks designed to bypass deepfake detection.
7. Future deployment should consider privacy, biometric protection, and compliance with data protection regulations.

These future developments may contribute to the creation of more reliable, scalable, and intelligent deepfake audio detection systems capable of supporting secure voice authentication in modern communication infrastructures.

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